

Introduction to Video

Video is a sequence of still images (frames) displayed rapidly to create the illusion of continuous motion. This phenomenon occurs because of **persistence of vision**, where the human eye briefly retains an image (about **1/16 second**) after it disappears. When images are shown quickly enough, the brain perceives smooth motion instead of separate pictures.

Definition

Video is a collection of sequential images (frames), synchronized with audio when required, that creates the perception of moving visual scenes.

1. Digital Video

Definition

Digital video is a sequence of digital images (frames) displayed rapidly to create the illusion of motion. Each frame is a still image, and when frames are displayed continuously (typically 24–60 frames per second), the human eye perceives smooth movement.

Digital video is the electronic representation of moving images where each frame is stored as binary data (0s and 1s). It can be processed, edited, compressed, transmitted, and displayed using computer systems.

Unlike analog video, digital video stores visual information numerically, making it more suitable for modern multimedia applications.

Components of Digital Video

- **Frames:** Individual images in a video.
- **Frame Rate (fps):** Number of frames displayed per second (e.g., 24 fps, 30 fps, 60 fps).
- **Resolution:** Number of pixels in each frame (e.g., 1280×720, 1920×1080).
- **Color Depth:** Number of bits used to represent colors.
- **Bit Rate:** Amount of data processed per second (kbps or Mbps).

Digital Video Components

Every digital video consists of several components.

A. Frames

A **frame** is a single still image in a video sequence.

Example:

A 10-second video recorded at **30 fps** contains:

$$10 \times 30 = 300 \text{ frames}$$

Each frame contributes to the illusion of continuous motion.

B. Pixels

A **pixel (Picture Element)** is the smallest addressable unit of a digital image.

For a resolution of:

$$1920 \times 1080$$

Total pixels:

$$1920 \times 1080 = 2,073,600$$

Approximately **2.07 megapixels**.

C. Resolution

Resolution specifies the number of pixels displayed horizontally and vertically.

Higher resolution generally provides greater detail but also increases storage and bandwidth requirements.

Resolution	Dimensions	Common Name
640 × 480	VGA	Standard Definition
1280 × 720	HD	High Definition
1920 × 1080	Full HD	Full High Definition
2560 × 1440	QHD	Quad HD
3840 × 2160	4K UHD	Ultra HD
7680 × 4320	8K UHD	Ultra High Definition

D. Frame Rate (Frames Per Second)

Frame rate indicates the number of frames displayed every second.

Formula

FPS=Number of Frames/Time

Common Frame Rates

FPS	Applications
24	Cinema
25	PAL television
30	NTSC/web video
50	Sports broadcasting
60	Gaming, sports, smooth motion
120+	Slow-motion capture and high-performance gaming

Higher frame rates produce smoother motion but require more processing power and storage.

E. Aspect Ratio

Aspect ratio is the proportional relationship between the width and height of a video.

Formula

Aspect Ratio=Width/Height

Common Aspect Ratios

Aspect Ratio	Applications
4:3	Older televisions and monitors
16:9	HDTV, YouTube, streaming
21:9	Cinema, ultrawide displays
1:1	Social media posts
9:16	Vertical smartphone videos

F. Bit Depth

Bit depth determines how many colors can be represented for each pixel.

Formula

$$\text{Colors} = 2^n$$

Where **n** is the number of bits.

Bit Depth	Possible Colors
8-bit	256 values per channel
24-bit RGB	About 16.7 million colors
30-bit RGB	About 1.07 billion colors

Higher bit depth allows smoother gradients and more accurate color reproduction.

G. Bit Rate

Bit rate is the amount of data processed or transmitted each second.

Unit

- kbps (kilobits per second)
- Mbps (megabits per second)

Higher bit rates usually improve quality but increase file size.

Example:

- 5 Mbps → Standard HD streaming
 - 25 Mbps → Typical 4K streaming
-

Characteristics of Digital Video

1. Stored as binary data.
2. High image quality.
3. Easy to edit and process.
4. Supports compression.
5. Efficient transmission over networks.
6. Can include synchronized audio, subtitles, and metadata.
7. Compatible with streaming platforms.

Analog Video vs Digital Video

Feature	Analog Video	Digital Video
Representation	Continuous electrical signal	Binary data
Noise	More susceptible	Less susceptible
Editing	Difficult	Easy
Storage	Magnetic tape	Hard disks, SSDs, cloud
Transmission	Signal quality degrades	Can be copied without cumulative quality loss
Compression	Limited	Extensive codec support
Cost	Older technology	Widely used today

Advantages of Digital Video

- High picture quality.
- Easy editing and post-production.
- Efficient compression reduces storage needs.
- Supports online streaming.
- Can be copied repeatedly with minimal quality degradation.
- Compatible with AI-based video analysis.

Limitations of Digital Video

- High-resolution videos require substantial storage.
- Editing and rendering demand powerful hardware.
- Large files require higher network bandwidth.
- Compression artifacts may appear when bit rates are too low.

Applications

- Online streaming (YouTube, Netflix)
- Video conferencing
- Digital television
- E-learning
- Entertainment and gaming

2. Video Signal Representation

Definition

Video signal representation is the process of representing visual information electronically so that it can be stored, transmitted, and displayed.

There are two major types:

A. Analog Video Signal

- Continuous electrical signal
- Used in traditional television systems
- Examples:
 - NTSC
 - PAL
 - SECAM

Characteristics

- Susceptible to noise
- Lower picture quality
- Difficult to edit

NTSC (National Television System Committee)

Definition

NTSC is the first widely adopted analog color television standard, developed by the **National Television System Committee** in the United States in **1953**. It was designed to provide color television while remaining compatible with existing black-and-white TVs.

Features

- **Frame Rate:** 30 frames per second (actually 29.97 fps)
- **Fields per Second:** 60
- **Resolution:** 525 scan lines
- **Frequency:** 60 Hz
- **Color Encoding:** Uses the YIQ color model

PAL (Phase Alternating Line)

Definition

PAL is an analog color television standard developed in **Germany** by Walter Bruch in **1963**. It improves color accuracy by automatically correcting color errors during transmission.

SECAM (Sequential Color with Memory)

Definition

SECAM is an analog color television standard developed in **France** in **1967**. Unlike NTSC and PAL, SECAM transmits color information sequentially and uses memory in the television receiver to reconstruct the full-color image.

B. Digital Video Signal

- Represents video as binary data (0s and 1s).
- Used in modern multimedia systems.

Digital Video Representation

Each video consists of:

```
Video
  ↓
Frames
  ↓
Pixels
  ↓
Binary Values
```

Example:

```
Video
Frame 1
Frame 2
Frame 3
...
Frame n
```

Each frame contains thousands or millions of pixels.

Color Representation

Each pixel is represented using RGB values.

Example:

Pixel RGB Value

Red (255,0,0)

Green (0,255,0)

Blue (0,0,255)

3. Computer Video Format

Definition

A computer video format specifies how video data is stored, compressed, and played on computer systems.

It includes:

- Video codec
 - Audio codec
 - File container
-

Common Video Formats

Format Extension		Features
MP4	.mp4	Most popular, high compression
AVI	.avi	High quality, larger file size
MOV	.mov	Apple QuickTime format
WMV	.wmv	Microsoft Windows Media
MKV	.mkv	Supports multiple audio/subtitle tracks
FLV	.flv	Adobe Flash Video
WebM	.webm	Used for web streaming

Video Compression Standards

Codec	Description
H.264	Widely used, efficient compression
H.265 (HEVC)	Better compression than H.264
VP9	Open-source codec by Google
AV1	Modern royalty-free codec

Factors Affecting Video Size

- Resolution
 - Frame rate
 - Color depth
 - Bit rate
 - Compression method
-

4. Computer-Based Animation

Definition

Computer-based animation is the process of creating moving images using computer software by displaying a sequence of images.

Animation creates the illusion of movement by rapidly displaying slightly different images.

Working Principle

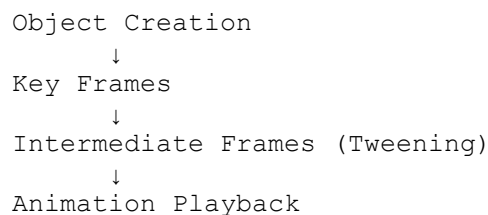
```
Object Creation
  ↓
Key Frames
  ↓
Intermediate Frames (Tweening)
  ↓
Animation Playback
```

Computer-Based Animation

Introduction

Computer-based animation is the process of creating the illusion of movement by displaying a sequence of images (frames) in rapid succession. Instead of drawing every frame manually, modern animation software allows animators to define the important positions of an object, and the computer automatically generates the remaining frames. This process saves time, reduces effort, and produces smooth, realistic animations.

The working principle of computer-based animation consists of four major stages:



1. Object Creation

Definition

Object creation is the first step in animation, where the animator designs or imports the objects that will appear in the animation. These objects can be simple shapes (circles, rectangles), text, images, characters, vehicles, or complex 3D models.

The object is the element that will be animated by changing its position, size, rotation, color, or other properties over time.

Activities Performed

- Designing the object
- Selecting colors and textures
- Defining object size and shape
- Positioning the object on the screen
- Setting initial properties (location, opacity, scale)

Example

Suppose we want to animate a **car** moving from the left side of the screen to the right.

Initially, the animator creates or imports the car model and places it at the starting position.

Time = 0 second



Common Software Used

- Adobe Animate
- Blender
- Autodesk Maya
- Unity
- Unreal Engine

2. Key Frames

Definition

A **key frame** is an important frame that defines the starting, ending, or significant position of an object during an animation. Instead of specifying every frame, the animator only defines these important moments.

The computer uses these key frames as reference points to generate the remaining frames automatically.

Purpose of Key Frames

Key frames specify changes such as:

- Position
- Rotation
- Size (Scaling)
- Color
- Opacity
- Shape

Example

Suppose a ball moves from the left side of the screen to the right in **5 seconds**.

Key Frame 1 (Start):

Time = 0 sec



Key Frame 2 (End):

Time = 5 sec



Only these two positions are defined by the animator.

Advantages

- Reduces manual work.
- Easy to modify the animation.
- Provides better control over motion.
- Speeds up animation development.

3. Intermediate Frames (Tweening)

Definition

Tweening (short for **in-betweening**) is the process in which the computer automatically generates the intermediate frames between two key frames.

Instead of drawing every frame manually, the animation software calculates the object's position, size, rotation, and other properties for each intermediate frame.

Working Process

Suppose an animation runs at **30 frames per second (fps)** and lasts **2 seconds**.

Total frames:

$$30 \times 2 = \mathbf{60 \text{ frames}}$$

If:

- Frame 1 is the starting key frame.
- Frame 60 is the ending key frame.

The computer automatically generates:

Frames 2–59.

Example

A ball moving horizontally:

Frame 1



Frame 15



Frame 30



Frame 45



Frame 60



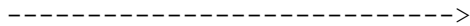
Only Frame 1 and Frame 60 are key frames. All other frames are created automatically.

Types of Tweening

A. Motion Tweening

Changes the object's position.

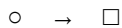
Example:



B. Shape Tweening

Changes one object into another.

Example:



C. Color Tweening

Gradually changes colors.

Example:

Red → Orange → Yellow → Green

D. Scale Tweening

Changes object size.

Example:

Small Circle → Medium Circle → Large Circle

E. Rotation Tweening

Rotates an object smoothly.

Example:

0° → 90° → 180° → 270°

Advantages of Tweening

- Produces smooth animation.
 - Saves significant time.
 - Requires fewer manually drawn frames.
 - Makes movement more natural and realistic.
-

4. Animation Playback

Definition

Animation playback is the final stage where all generated frames are displayed continuously at a specified frame rate to create the illusion of motion.

The animation software rapidly displays one frame after another, making static images appear as moving objects.

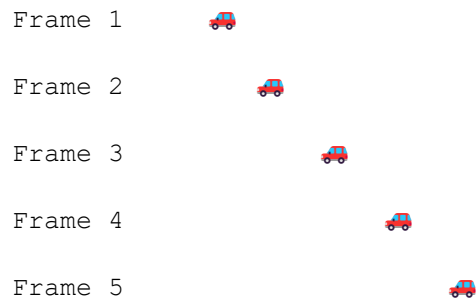
Frame Rate

Frame rate determines how many frames are displayed every second.

Frame Rate	Application
24 fps	Movies
25 fps	Television (PAL)
30 fps	Online videos
60 fps	Gaming and sports

Higher frame rates result in smoother animations but require more processing power and storage.

Playback Example



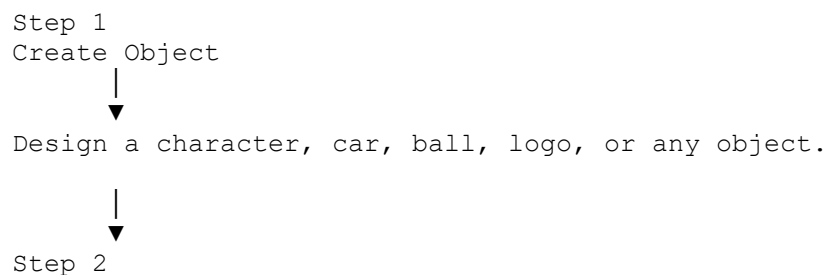
Since these frames are displayed very quickly, the car appears to move smoothly across the screen.

During Playback

The computer performs the following tasks:

- Displays frames in sequence.
- Maintains the specified frame rate.
- Synchronizes audio with video.
- Renders lighting, textures, and visual effects (for 3D animation).
- Handles user interaction in interactive applications (e.g., games).

Complete Animation Process



Create Key Frames



Define the object's important positions and transformations.



Step 3
Tweening



The computer automatically generates intermediate frames.



Step 4
Playback



Frames are displayed rapidly to create smooth motion.

Real-Life Example

Suppose you are creating an animation of a **bird flying across the sky**.

Step 1: Object Creation

Create or import the bird image.



Step 2: Key Frames

- Key Frame 1: Bird on the left side.
- Key Frame 2: Bird on the right side.

Time = 0 sec



Time = 5 sec



Step 3: Tweening

The software generates all intermediate positions automatically.



Step 4: Playback

The generated frames are displayed at **30 fps**, making the bird appear to fly smoothly across the screen.

Advantages of This Animation Process

- Saves time and effort.
 - Produces smooth and realistic movement.
 - Simplifies editing by modifying only key frames.
 - Reduces manual drawing of every frame.
 - Supports complex effects such as scaling, rotation, fading, and color changes.
-

Applications

- Animated movies and cartoons
 - Video games
 - Mobile applications
 - Website animations
 - Educational simulations
 - Scientific visualization
 - Advertising and marketing
 - User interface (UI) animations
-

Summary

The **working principle of computer-based animation** involves four sequential stages:

1. **Object Creation:** Create or import the object to be animated.
2. **Key Frames:** Define the important positions or transformations of the object.
3. **Intermediate Frames (Tweening):** Automatically generate the in-between frames to create smooth transitions.
4. **Animation Playback:** Display all frames rapidly (e.g., 24–60 fps) to produce the illusion of continuous motion.

Types of Computer Animation

A. 2D Animation

Objects move in two dimensions (X and Y).

Examples:

- Cartoons
 - Educational videos
-

B. 3D Animation

Objects move in three dimensions (X, Y, and Z).

Examples:

- Animated movies
 - Video games
 - Product visualization
-

C. Motion Graphics

Animated text, logos, and graphical elements.

Applications:

- Advertisements
 - News broadcasts
 - Title sequences
-

D. Stop Motion Animation

Creates animation by photographing physical objects frame by frame.

Animation Techniques

1. Frame-by-Frame Animation

Each frame is drawn individually.

2. Keyframe Animation

Animator specifies important frames, and the computer generates intermediate frames (tweening).

3. Procedural Animation

Movement is generated using mathematical algorithms or physics simulations.

Applications

- Gaming
 - Movies
 - Medical visualization
 - Education
 - Scientific simulations
 - Advertising
-

5. Animation Language

Definition

An animation language is a programming or scripting language used to define, control, and create animations.

It specifies:

- Object movement
 - Rotation
 - Scaling
 - Timing
 - Color changes
 - Interaction
-

Basic Components of Animation Language

1. Objects

Elements to be animated.

Example:

- Circle
 - Car
 - Human character
-

2. Motion Commands

Examples:

- Move
 - Rotate
 - Scale
 - Translate
 - Zoom
-

3. Timing Control

Specifies:

- Start time
- End time
- Speed
- Delay

Example:

```
Move Object  
Start = 2 sec  
End = 5 sec
```

4. Transformation

Changes object properties:

- Position
- Size

- Rotation
- Color

Common Animation Languages and Technologies

Language/Technology	Purpose
JavaScript	Web animations
CSS Animations	Web page effects
SVG Animation	Vector graphics animation
Adobe Animate (ActionScript)	Interactive animations
Python (Pygame, Manim)	Educational and scientific animations
Unity C#	Game animations
Unreal Engine Blueprints/C++	3D game animations

Advantages of Animation Languages

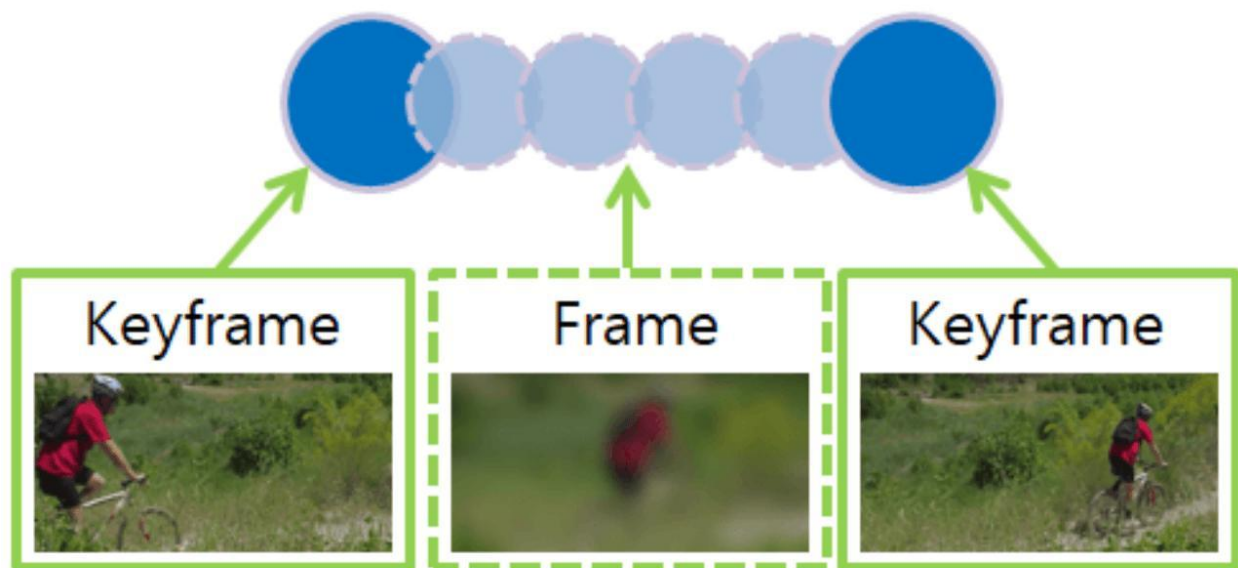
- Automate animation creation
- Reusable animation scripts
- Interactive animations
- Faster development
- Better control over movement and timing

Timeline and Frame-Based Animation

Learning Objectives

After studying this chapter, students will be able to:

- Define **Timeline and Frame-Based Animation**.
 - Explain the concept of **Timeline**.
 - Describe **Frames** and **Keyframes**.
 - Calculate and interpret **Frame Rate (FPS)**.
 - Understand **Onion Skinning**.
 - Create simple frame-by-frame animations.
 - Compare frame-based animation with tween-based animation.
 - Explain practical applications in multimedia, gaming, education, and film.
-



Comparison: Frame-Based vs Tween-Based Animation

Feature	Frame-Based Animation	Tween-Based Animation
Frame Creation	Every frame is drawn manually	Only keyframes are created; software generates intermediate frames
Time Required	High	Lower
Flexibility	Very high	Moderate
Best For	Character animation, cartoons, artistic motion	Logos, UI effects, object movement
Software Automation	Minimal	Extensive

Summary

Concept	Description
Timeline	Organizes frames over time and controls animation duration
Frame	A single still image in an animation sequence
Keyframe	A frame marking a significant change in an object's properties
Frame Rate (FPS)	Number of frames displayed each second
Onion Skinning	Displays neighboring frames transparently to assist smooth animation
Frame-Based Animation	Animation method where each frame is created individually

1. Introduction to Timeline and Frame-Based Animation

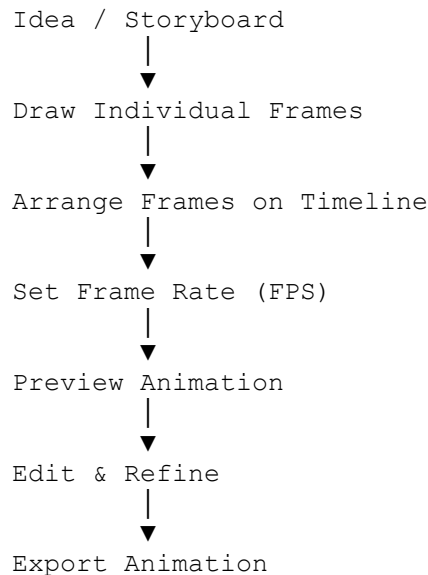
Animation is the illusion of movement created by displaying a sequence of images rapidly. In **frame-based animation**, every frame is created or edited individually. These frames are arranged on a **timeline**, which controls when each frame appears.

Unlike **tween-based animation**, where software automatically generates intermediate frames, **frame-based animation** requires the animator to design each frame manually.

Definition

Timeline and Frame-Based Animation is an animation technique in which every frame is individually created and organized on a timeline to produce continuous motion when displayed in sequence.

Animation Workflow



Stage 1: Idea / Storyboard

Definition

The **Idea** stage is the creative planning phase where the animator decides **what story or message the animation will communicate**. After developing the idea, it is organized into a **storyboard**, which is a series of sketches showing the sequence of scenes.

A storyboard acts as the **blueprint** of the animation before production begins.

Objectives

- Define the animation's purpose.
- Plan the storyline.

- Organize scenes logically.
 - Reduce production errors.
 - Estimate production time and resources.
-

Storyboard Components

A storyboard typically includes:

- Scene number
 - Sketch or illustration
 - Description of the action
 - Camera movement
 - Dialogue or narration
 - Sound effects
 - Estimated duration
-

Example Storyboard

Scene	Sketch	Description	Duration
1	Student enters classroom	Introduction	3 sec
2	Teacher starts lecture	Learning begins	5 sec
3	Students use computers	Practical session	6 sec
4	Class ends	Conclusion	2 sec

Advantages

- Improves planning.
 - Saves production time.
 - Helps the team visualize the final animation.
 - Identifies mistakes before production.
-

Real-Life Example

Before producing an animated movie, studios first prepare hundreds or even thousands of storyboard sketches to visualize the complete story before creating the actual animation.

Stage 2: Draw Individual Frames

Definition

After planning, the animator creates each **individual frame**. A frame is one still image, and many frames displayed rapidly create the illusion of movement.

In **frame-by-frame animation**, every frame is drawn manually.

Process

1. Draw the first frame.
 2. Draw the next frame with a slight change.
 3. Continue until the movement is complete.
-

Example: Moving Ball

Frame 1

O

Frame 2

O

Frame 3

O

Frame 4

O

Each frame moves the ball slightly to the right.

Best Practices

- Keep object proportions consistent.
 - Make smooth, gradual movements.
 - Use onion skinning to compare adjacent frames.
-

Applications

- Cartoon animation
 - Character animation
 - Hand-drawn animation
 - Traditional 2D animation
-

Stage 3: Arrange Frames on Timeline

Definition

After creating the frames, they are placed on the **timeline** in chronological order. The timeline determines **when** each frame is displayed.

The timeline acts as the control panel for the animation.

Timeline Example

Time →

Frame Number

1 2 3 4 5 6 7 8

■ ■ ■ ■ ■ ■ ■ ■

Each block represents one frame.

Timeline Functions

- Organizes animation.
- Controls timing.
- Synchronizes audio.
- Supports multiple layers.
- Allows insertion of keyframes.

Multi-Layer Timeline

Background



Character



Text



Different elements can be animated independently.

Advantages

- Easy editing.
- Better synchronization.
- Precise timing control.
- Organized workflow.

Stage 4: Set Frame Rate (FPS)

Definition

The **Frame Rate** specifies how many frames are displayed each second.

It determines the smoothness of the animation.

Formula

$\text{FPS} = \frac{\text{Number of Frames}}{\text{Time (seconds)}}$
 $\text{FPS} = \text{Number of Frames} / \text{Time (seconds)}$

Example 1

Animation Duration = **5 seconds**

Frames = **150**

$\text{FPS} = 150 / 5 = 30 \text{ FPS}$

Therefore, the animation runs at **30 FPS**.

Example 2

Movie Duration = **120 seconds**

$\text{FPS} = 24$

Total Frames

$24 \times 120 = 2880$

Common Frame Rates

FPS	Applications
12	Simple web animations
24	Cinema
25	PAL television
30	Online videos and TV
60	Gaming and sports
120	Slow-motion recording

Effect of FPS

Low FPS	High FPS
Choppy movement	Smooth movement

Low FPS	High FPS
Smaller file size	Larger file size
Lower processing requirements	Higher processing requirements

Stage 5: Preview Animation

Definition

The preview stage allows the animator to **play the animation before final production** to check whether everything works correctly.

This stage is similar to testing software before release.

What Is Checked?

- Smoothness of movement
 - Object positions
 - Timing
 - Audio synchronization
 - Camera movement
 - Lighting and colors
 - Animation speed
-

Example

If a character appears to "jump" unexpectedly between frames, the animator can detect the issue during preview instead of after exporting the final animation.

Advantages

- Detects mistakes early.
- Saves rendering time.
- Improves overall quality.
- Ensures synchronization.

Stage 6: Edit & Refine

Definition

After previewing, the animation is improved by correcting errors and enhancing its quality.

This stage is often repeated several times until the animation meets the desired standard.

Editing Activities

A. Correct Drawing Errors

- Fix incorrect shapes.
 - Maintain character proportions.
-

B. Improve Timing

Adjust frame spacing to make movement appear more natural.

C. Modify Colors

Improve brightness, contrast, and color consistency.

D. Add Special Effects

Examples:

- Shadows
- Fire
- Smoke
- Rain
- Motion blur
- Lighting effects

E. Synchronize Audio

Align:

- Background music
- Voice narration
- Sound effects

with the animation.

F. Optimize Performance

- Remove unnecessary frames.
 - Compress assets.
 - Reduce rendering time.
-

Example

During preview, a walking character appears to move too quickly.

The animator increases the number of intermediate frames, making the walking motion smoother and more realistic.

Stage 7: Export Animation

Definition

The final animation is converted into a standard file format so it can be viewed, shared, or published.

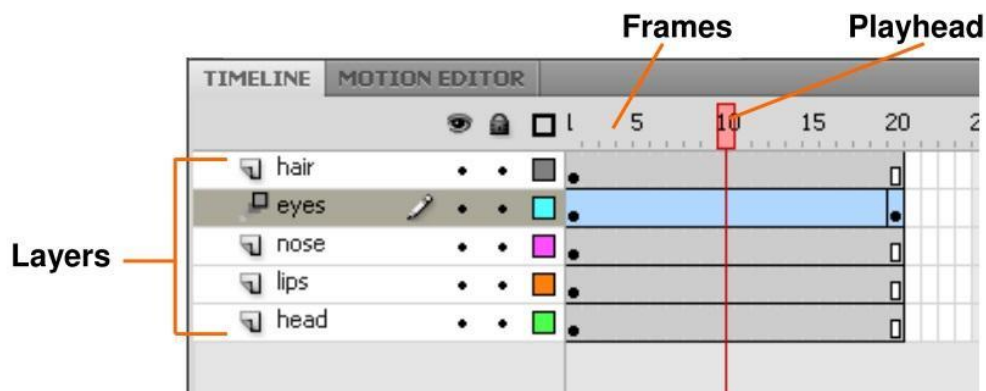
Common Export Formats

Format	Applications
MP4	YouTube, websites, social media
MOV	Professional video editing
AVI	Windows multimedia
GIF	Short looping animations
WebM	Web applications
SWF (legacy)	Older web animations

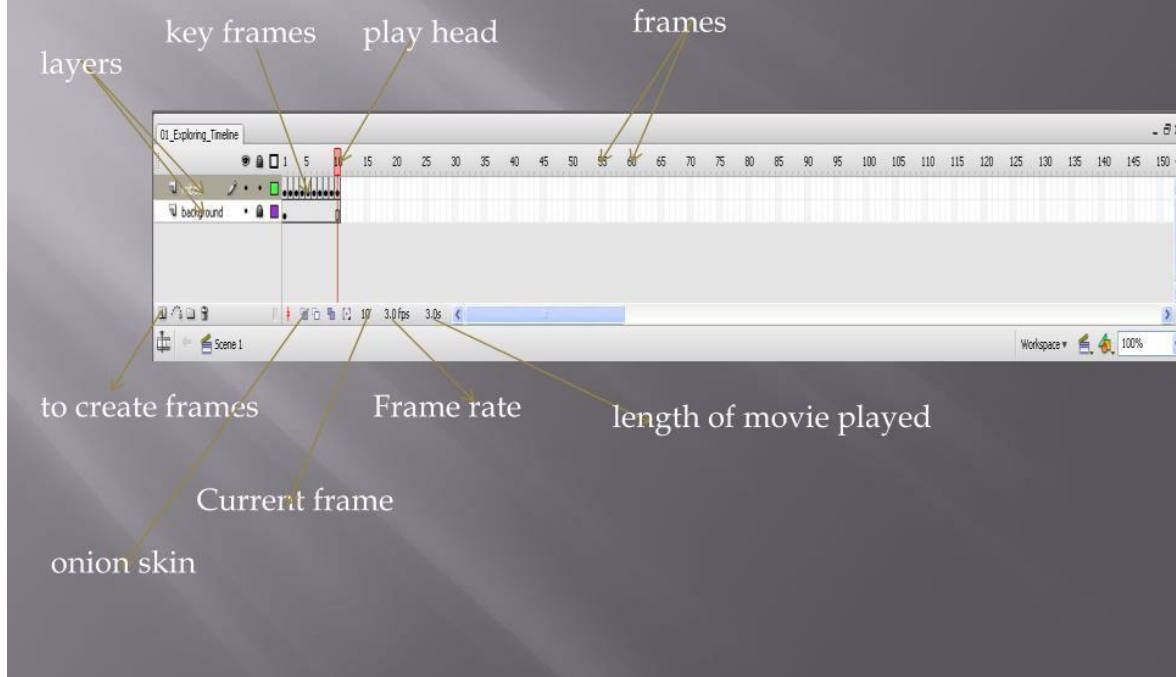
2. Timeline

Timeline

You can organize and control the content of a movie over time.



Layout of Timeline



5

Definition

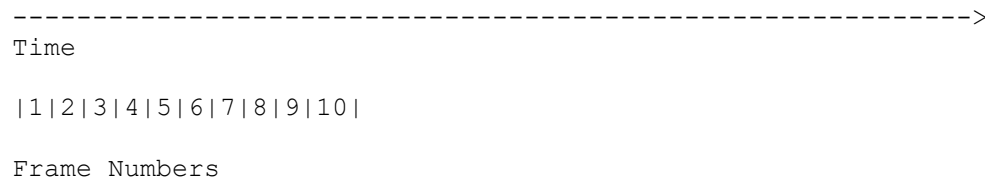
A **Timeline** is the main workspace used to organize frames over time. It determines **when** objects appear, move, disappear, or change.

Think of a timeline as the **time axis** of an animation.

Characteristics of Timeline

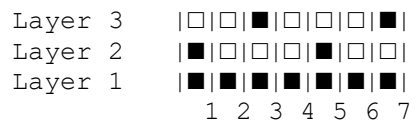
- Organizes animation from beginning to end.
- Represents time from left to right.
- Contains frames and keyframes.
- Supports multiple layers.
- Controls animation duration.
- Synchronizes audio and visual elements.

Timeline Components



Each box represents one frame.

Timeline with Layers



Where:

- □ = Empty Frame
 - ■ = Frame containing content
-

Advantages of Timeline

- Easy animation management.
 - Supports multiple objects independently.
 - Allows precise timing.
 - Simplifies synchronization.
 - Makes editing straightforward.
-

Applications

- Cartoon production
- Educational videos
- Motion graphics
- Video games
- Web animation

3. Frames

Definition

A **Frame** is one individual image displayed in an animation sequence.

When multiple frames are shown rapidly, they create the illusion of movement.

Example

Suppose a ball moves from left to right.

Frame 1

O

Frame 2

O

Frame 3

O

Frame 4

O

Each frame shows the ball in a slightly different position.

Types of Frames

A. Empty Frame

Contains no object.

[]

B. Static Frame

Object remains unchanged.

[Ball]

C. Content Frame

Contains graphics, text, or animation elements.

D. Blank Keyframe

Used to begin a new animation sequence.

Frame Number

Frames are numbered sequentially.

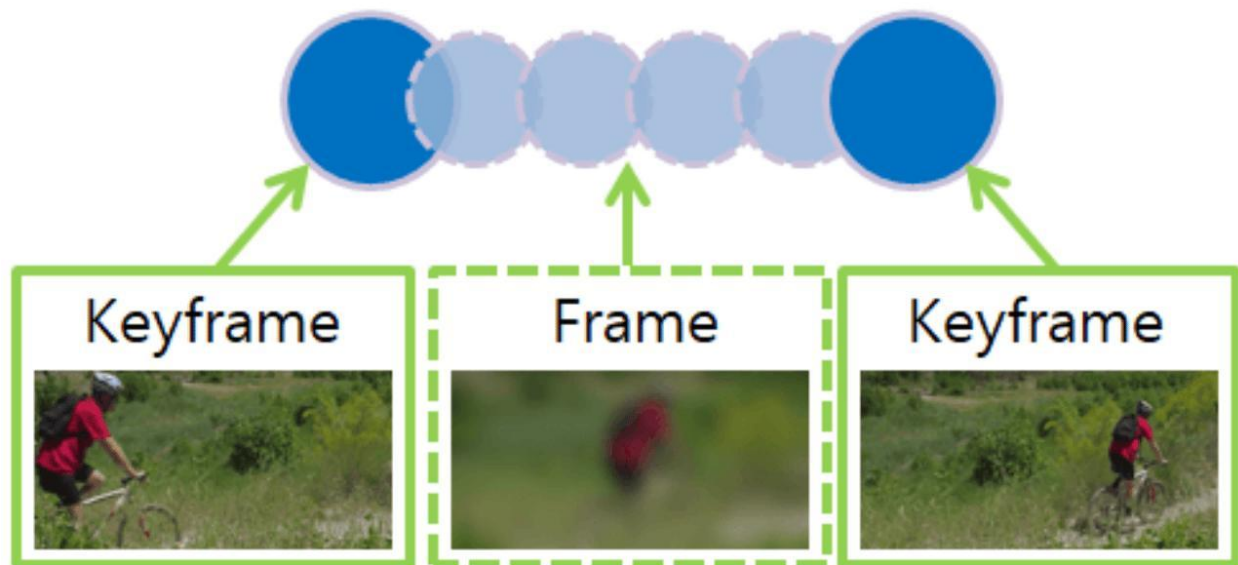
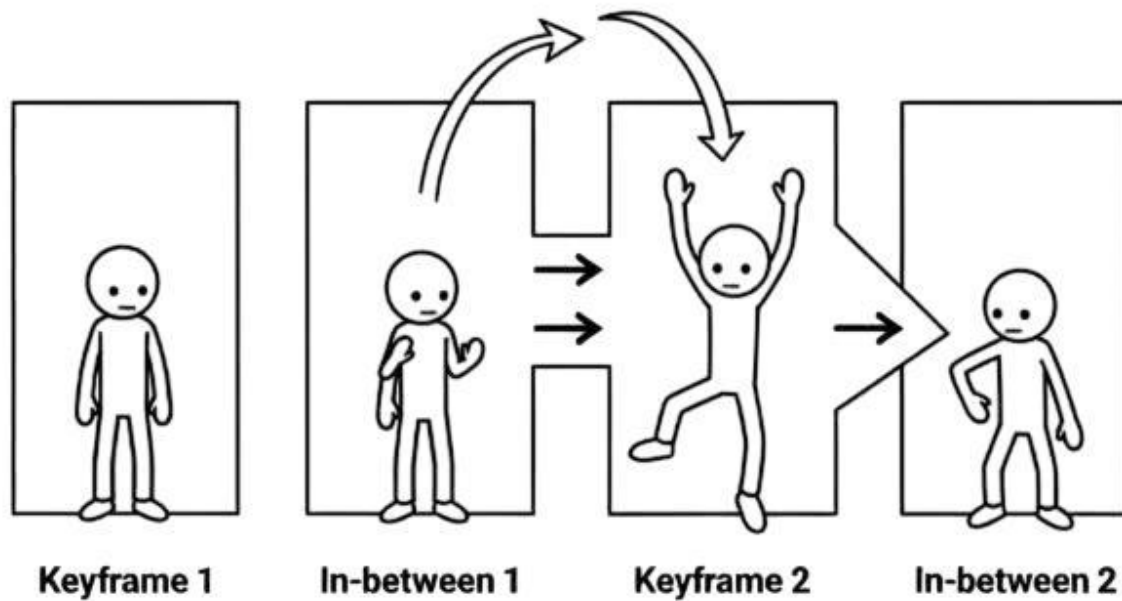
Example:

1 2 3 4 5 6 7 8 9 10

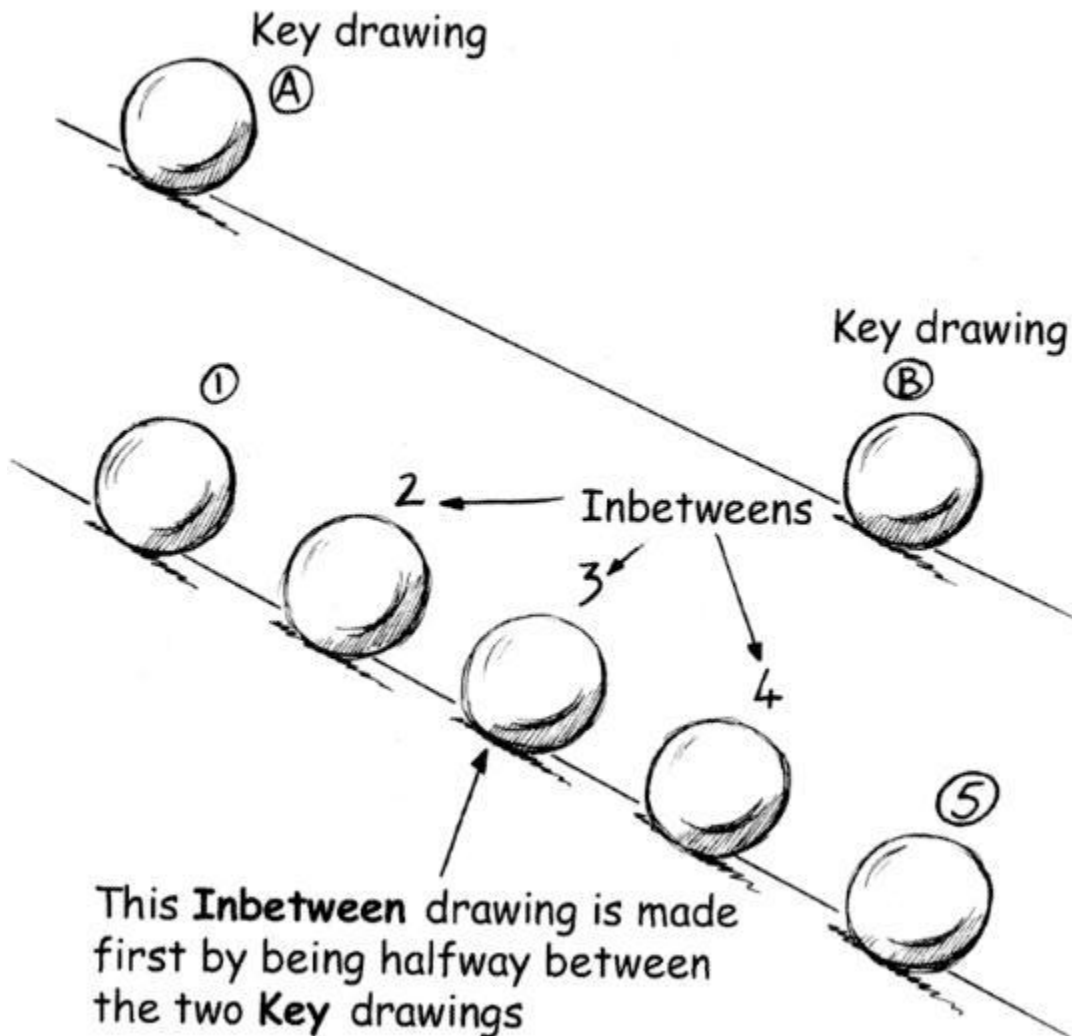
4. Keyframes

The Keyframe Technique:

DEFINING KEY POSES AND IN-BETWEEN FRAMES



Key Drawings & Inbetweens



6

Definition

A **Keyframe** is a special frame where an important change occurs.

Examples of changes include:

- Position
- Rotation

- Scale
- Opacity
- Color
- Shape

The animation software uses keyframes to define major states of an object.

Example

A ball moves across the screen.

Frame 1



Keyframe

Frame 15



Keyframe

Frame 30



Keyframe

Each keyframe records a significant position.

Types of Keyframes

Position Keyframe

Changes object location.

Rotation Keyframe

Changes object angle.

Scale Keyframe

Changes object size.

Color Keyframe

Changes object color.

Opacity Keyframe

Changes transparency.

Advantages

- Reduces repetitive work.
 - Makes editing easier.
 - Provides precise control.
 - Essential for complex animations.
-

5. Frame Rate (FPS)

Definition

Frame Rate is the number of frames displayed every second.

Unit

Frames Per Second (FPS)

Formula

$\text{FPS} = \text{Number of Frames} / \text{Time (seconds)}$

Example 1

Animation Duration = 5 seconds

Frames = 150

$\text{FPS} = 150 / 5 = 30\text{FPS}$

Frame Rate = **30 FPS**

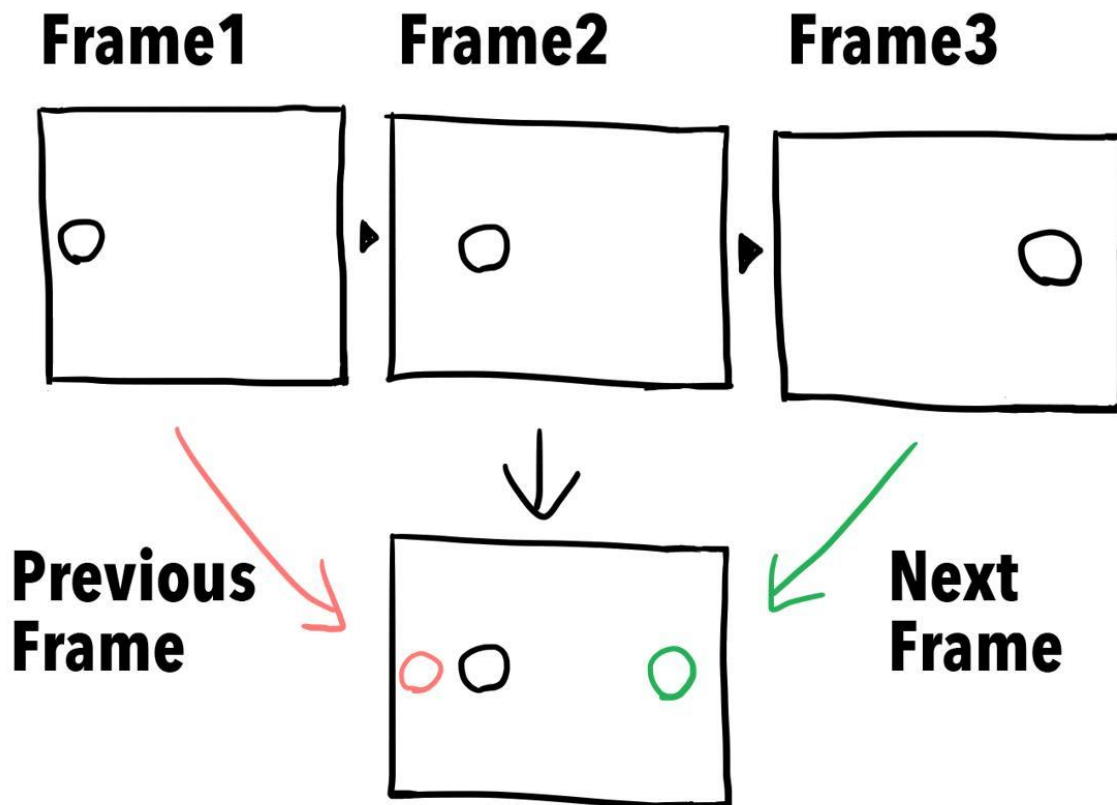
Standard Frame Rates

FPS	Applications
12	Simple web animations
24	Cinema
25	PAL television
30	Television and online videos
60	Gaming and sports
120	High-speed recording and slow motion

Effect of Frame Rate

Low FPS	High FPS
Choppy motion	Smooth motion
Less storage	More storage
Lower CPU usage	Higher CPU usage

6. Onion Skinning



7. Practical Example

Example 1: Bouncing Ball Animation

Frame 1

Ball at the top.

○

Frame 2

Ball moving downward.

○

Frame 3

Ball reaches the ground.

○

Frame 4

Ball bounces upward.

○

Frame 5

Ball returns to the top.

○

When these frames are played at **24–30 FPS**, the ball appears to bounce naturally.

Example 2: Walking Character

Frame 1

Left foot forward.

Frame 2

Standing position.

Frame 3

Right foot forward.

Frame 4

Standing position.

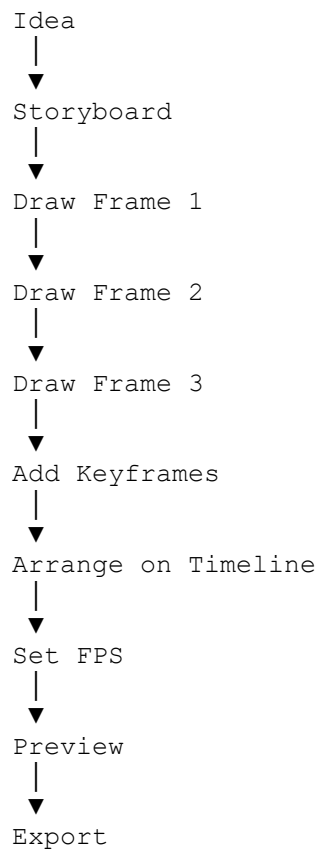
Repeating these frames creates a continuous walking cycle.

Example 3: Traffic Signal Animation

- Frame 1: Red light ON
- Frame 2: Yellow light ON
- Frame 3: Green light ON

Looping the sequence simulates a working traffic signal.

Frame-Based Animation Process



Advantages of Timeline and Frame-Based Animation

- Precise control over every frame.
 - Excellent for complex character animation.
 - Produces expressive and detailed movement.
 - Ideal for cartoons and artistic animation.
 - Easy to synchronize with audio.
-

Disadvantages

- Time-consuming because each frame is created individually.
 - Requires skilled animators.
 - Large projects involve many frames.
 - More editing effort than tween-based animation.
 - Increased storage and rendering time for lengthy animations.
-

Applications

Field	Application
Film Industry	Animated movies and visual effects
Television	Cartoons and advertisements
Education	Interactive learning materials
Gaming	Character and sprite animation
Medicine	Surgical and anatomical visualization
Engineering	Simulation of machines and processes
Scientific Research	Visualization of physical phenomena
Mobile Applications	User interface animations

Part I: Methods of Controlling Animation

What is Animation Control?

Animation control refers to the techniques used to determine **how**, **when**, and **where** an object moves, changes, or interacts during an animation.

Animation control manages:

- Position
- Rotation
- Scale
- Speed
- Direction
- Color
- Opacity
- User interaction

Comparison of Animation Control Methods

Method	Controlled By	Applications
Frame-by-frame Animator		Cartoons
Keyframe	Animator + Software	Motion graphics
Tween	Software	Advertisements
Script	Programming	Games
Event	User interaction	Interactive media
Time	System clock	Simulations
Physics	Physical equations	Engineering, games
AI	Machine learning	Smart animation

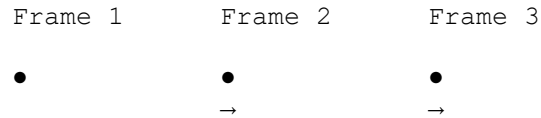
Types of Animation Control

1. Frame-by-Frame Control

Definition

Each frame is created manually by the animator.

Every movement is explicitly defined.



Characteristics

- Maximum control
- High-quality animation
- Time-consuming
- Suitable for cartoons

Applications

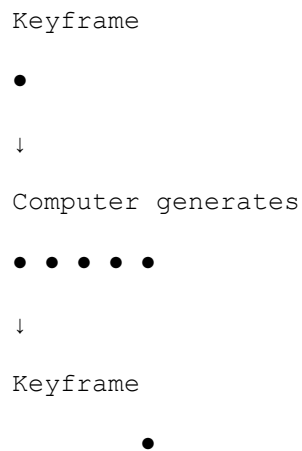
- Disney animation
- Anime
- Traditional cartoons

2. Keyframe-Based Control

Definition

The animator specifies only important frames (**keyframes**).

The computer automatically generates intermediate frames.



Advantages

- Faster production
- Smooth movement

- Easy editing

Example

A car moves from left to right.

- Frame 1 → Left
- Frame 60 → Right

The software generates Frames 2–59.

3. Tween-Based Control

Tween means "**in-between.**"

The software automatically creates intermediate frames.

Types

Motion Tween

Object changes position.

Example:

Logo slides across the screen.

Shape Tween

Object changes shape.

Example:

Circle

↓

Square

Color Tween

Changes color gradually.

Example:

Blue



Green

Opacity Tween

Changes transparency.

Example:

Invisible



Visible

4. Script-Based Control

Animation is controlled using programming languages.

Examples:

- JavaScript
- Python
- C#
- Unity Script

5. Event-Based Control

Animation starts after an event.

Events include:

- Mouse click
- Keyboard press
- Touch
- Voice command

6. Time-Based Control

Animation depends on elapsed time rather than frame count.

Example

Object moves every 0.5 seconds.

Advantages

- Consistent speed
 - Independent of computer performance
-

7. Physics-Based Control

Uses physical laws.

Examples

- Gravity
- Collision
- Wind
- Friction

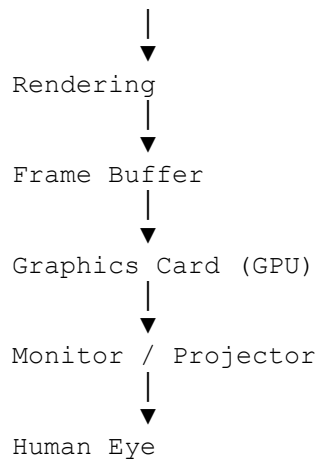
Part II: Display of Animation

Definition

Animation display is the process of presenting rendered animation on a display device.

Animation Display Process

Animation Frames



Step 1: Rendering

Rendering converts animation data into visible images.

Two methods

Real-Time Rendering

Produces images instantly.

Applications

- Games
- VR
- AR

Offline Rendering

Produces very high-quality images.

Applications

- Pixar movies
- Animated films

Step 2: Frame Buffer

A **frame buffer** is a memory area that temporarily stores rendered frames before they are displayed.

Benefits

- Smooth playback
- Reduces flickering
- Improves quality

Step 3: Double Buffering

Two buffers are used.

Buffer A

(Display)

Buffer B

(Render)

↓

Swap

Buffer B

(Display)

Buffer A

(Render)

Advantages

- Eliminates flickering
- Smooth animation

Triple Buffering

Uses three buffers.

Advantages

- Better performance
 - Higher FPS
 - Reduced screen tearing
-

Step 4: Graphics Processing Unit (GPU)

GPU responsibilities

- Render graphics
- Lighting
- Shadows
- Texture mapping
- Particle effects

Examples

- NVIDIA RTX
 - AMD Radeon
 - Intel Arc
-

Step 5: Display Devices

Animation can be displayed on

- LCD
 - LED
 - OLED
 - Smartphone
 - Projector
 - VR Headset
 - AR Glasses
-

Display Technologies

Technology	Characteristics
LCD	Affordable, widely used
LED	Brighter than LCD

Technology	Characteristics
OLED	Excellent contrast, deep blacks
VR Display	Immersive 3D environment
AR Display	Overlays digital objects on the real world

Factors Affecting Display Quality

- Resolution
 - FPS
 - Brightness
 - Contrast
 - Refresh Rate
 - Color Accuracy
 - GPU Performance
-

Part III: Transmission of Animation

Definition

Animation transmission is the process of delivering animation from one computer or server to another through a communication network.

Transmission Process

Animation

↓

Compression

↓

Encoding

↓

Internet

↓

Streaming

↓

Decoding

↓

Display

Step 1: Compression

Animation files are compressed to reduce size.

Benefits

- Faster download
- Less storage
- Lower bandwidth

Popular codecs

- H.264
 - H.265 (HEVC)
 - AV1
 - VP9
-

Step 2: Encoding

Encoding converts animation into a standard format.

Common formats

- MP4
 - WebM
 - MOV
 - AVI
-

Step 3: Transmission

Animation is transmitted through

- LAN
- Wi-Fi
- Internet
- Mobile Network
- Satellite

Step 4: Streaming

Streaming allows users to watch without downloading the entire file.

Types

Live Streaming

Example

Sports broadcasting

Video on Demand

Examples

- Netflix
- YouTube

Step 5: Buffering

Temporary storage before playback.

Benefits

- Smooth playback
 - Reduces interruptions
-

Step 6: Decoding

The receiver converts compressed animation back into playable frames.

Network Protocols

Protocol	Purpose
HTTP	Web delivery
HTTPS	Secure transmission
RTP	Real-time multimedia
RTSP	Streaming control
WebRTC	Real-time communication
TCP	Reliable delivery
UDP	Low-latency transmission

Animation Delivery Methods

Progressive Download

Downloads continuously while playing.

Streaming

Playback starts almost immediately.

Cloud Delivery

Animation is stored in cloud servers.

Examples

- YouTube
- Vimeo

- Google Drive